



Write the program

Name: _____

Date: _____



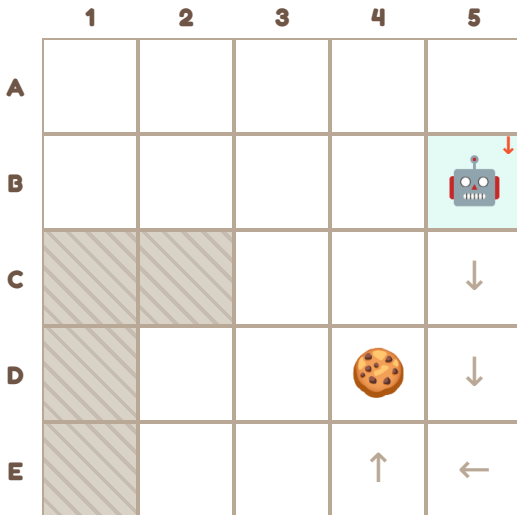
Robo starts on the green square. The small red arrow shows which way Robo is **facing**. Follow the grey arrows to the cookie crumbs and write one command in each box.

↑ go forward 1 square

← turn left (spin, don't move)

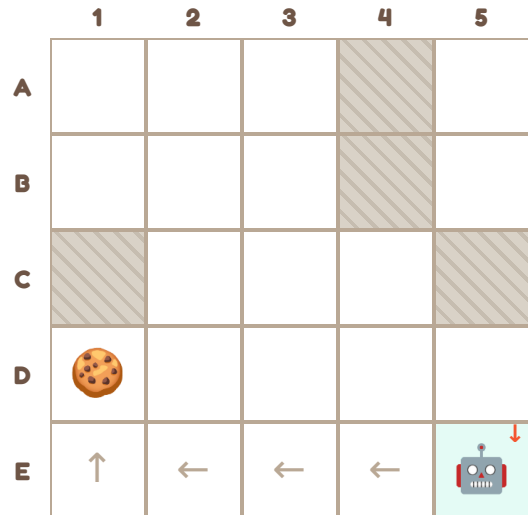
→ turn right (spin, don't move)

1



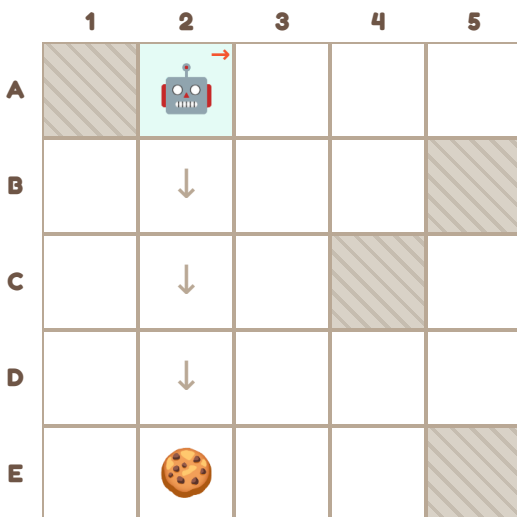
1	2	3	4	5	6	7

2



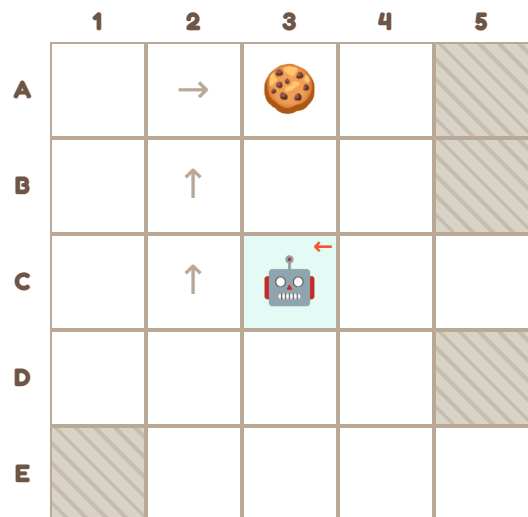
1	2	3	4	5	6	7

3



1	2	3	4	5

4



1	2	3	4	5	6



Where does Robo end up?

Name: _____

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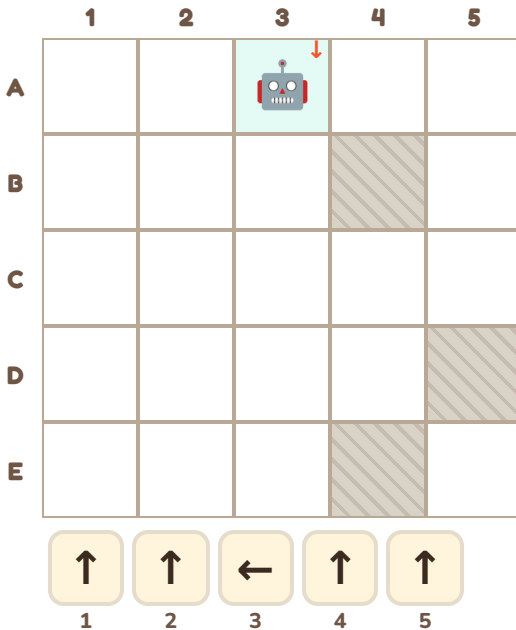
Read the program and move your finger square by square. **Draw a star** ★ in the square where Robo stops, and lightly colour every square Robo drove over.

↑ go forward 1 square

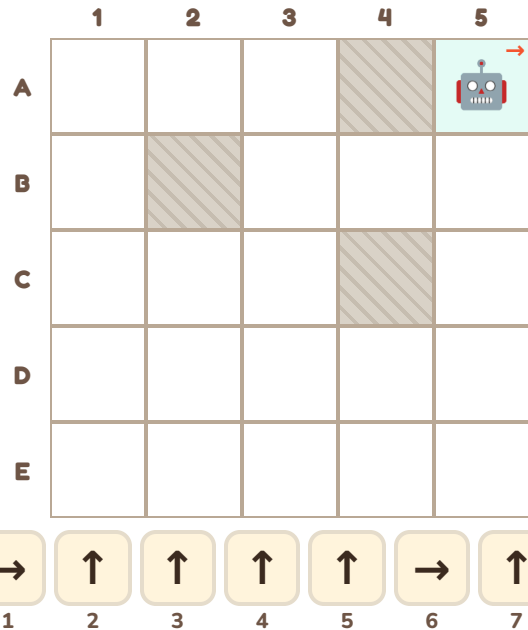
← turn left (spin, don't move)

→ turn right (spin, don't move)

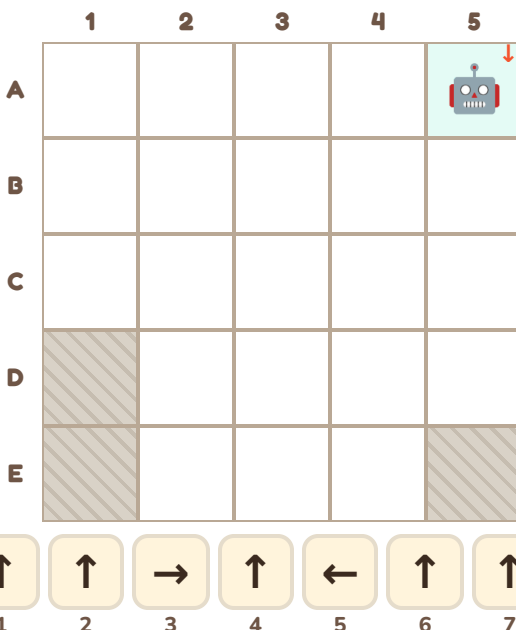
1



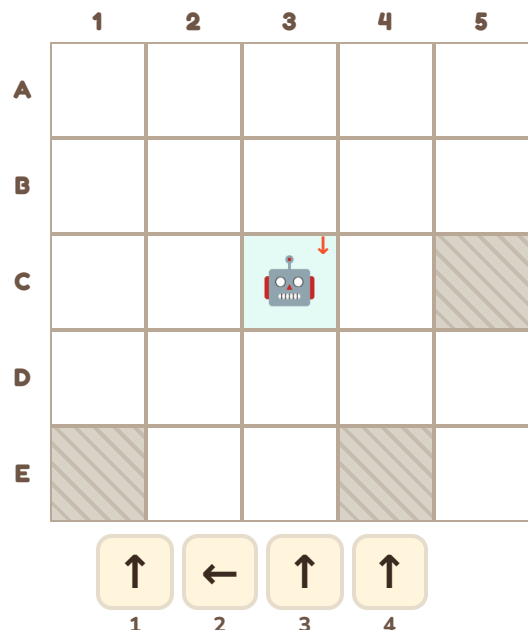
2



3



4





Find the bug

Name: _____

Date: _____



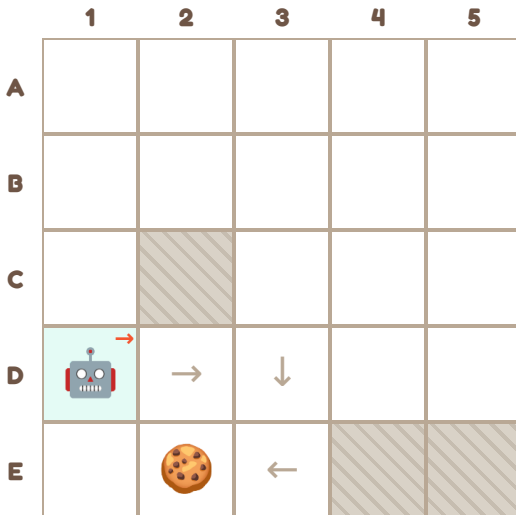
Each program has **one wrong command**, so Robo never reaches the crumbs. Follow the grey route, **circle the wrong command**, and write the correct one in the box under it.

↑ go forward 1 square

← turn left (spin, don't move)

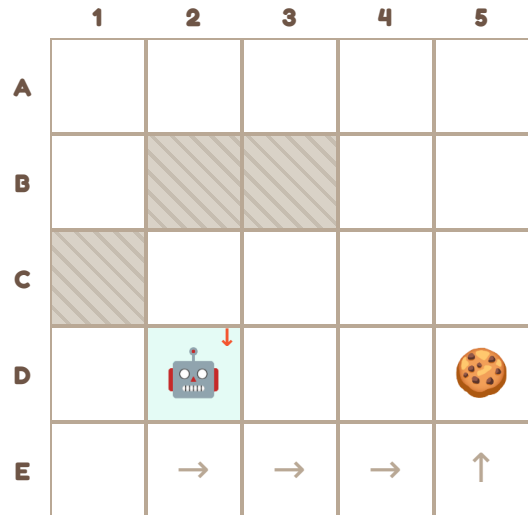
→ turn right (spin, don't move)

1



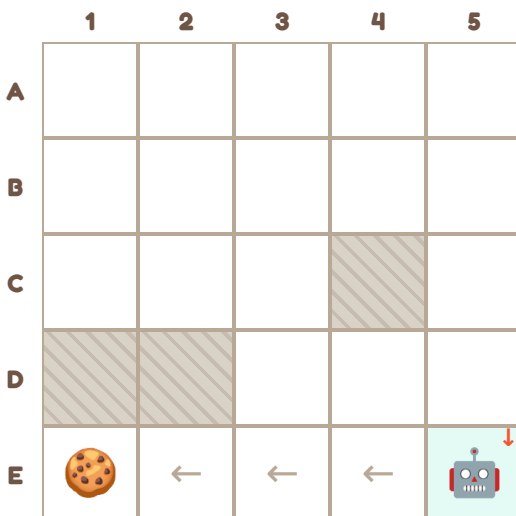
↑	↑	→	↑	→	→

2



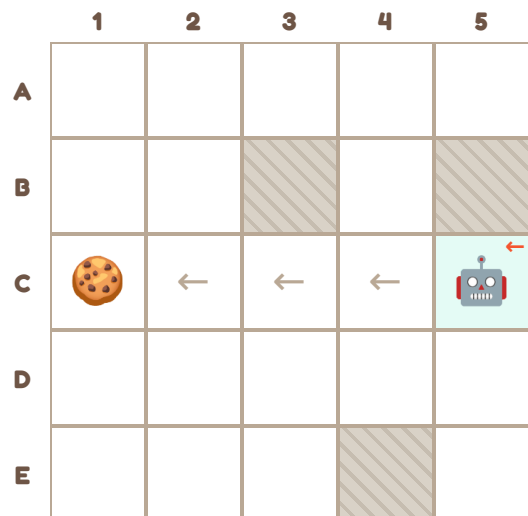
↑	←	↑	←	↑	←	↑

3



→	←	↑	↑	↑

4



→	↑	↑	↑



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A **repeat block** runs the same commands again and again. Look for the pattern that repeats. Write how many times, and the commands inside the block.

↑ go forward 1 square

← turn left (spin, don't move)

→ turn right (spin, don't move)

1

Robo's long program:



Write it as a repeat block:

Repeat times

2

Robo's long program:



Write it as a repeat block:

Repeat times

3

Robo's long program:



Write it as a repeat block:

Repeat times

4

Robo's repeat block:

Repeat **2** times

Write out the long program:



5

Robo's repeat block:

Repeat **3** times

Write out the long program:



6

Robo's repeat block:

Repeat **3** times

Write out the long program:



**Page 1: Write the program**

1. ↑ ↑ ↑ → ↑ → ↑ (7 commands, ends at D4)
2. → ↑ ↑ ↑ ↑ → ↑ (7 commands, ends at D1)
3. → ↑ ↑ ↑ ↑ (5 commands, ends at E2)
4. ↑ → ↑ ↑ → ↑ (6 commands, ends at A3)

Page 2: Where does Robo end up?

1. ends at **C5**, facing →; drives over A3 → B3 → C3 → C4 → C5
2. ends at **E4**, facing ←; drives over A5 → B5 → C5 → D5 → E5 → E4
3. ends at **E4**, facing ↓; drives over A5 → B5 → C5 → C4 → D4 → E4
4. ends at **D5**, facing →; drives over C3 → D3 → D4 → D5

Page 3: Find the bug

1. command **6** is wrong: → should be ↑ (correct program ↑ ↑ → ↑ → ↑)
2. command **4** is wrong: ← should be ↑ (correct program ↑ ← ↑ ↑ ↑ ← ↑)
3. command **2** is wrong: ← should be ↑ (correct program → ↑ ↑ ↑ ↑)
4. command **1** is wrong: → should be ↑ (correct program ↑ ↑ ↑ ↑)

Page 4: Repeat blocks

1. Repeat **4** times [↑]
2. Repeat **3** times [↑ →]
3. Repeat **3** times [↑ ←]
4. ↑ ↑ ← ↑ ↑ ← (6 commands)
5. → ↑ → ↑ → ↑ (6 commands)
6. ↑ ← ↑ ← ↑ ← (6 commands)